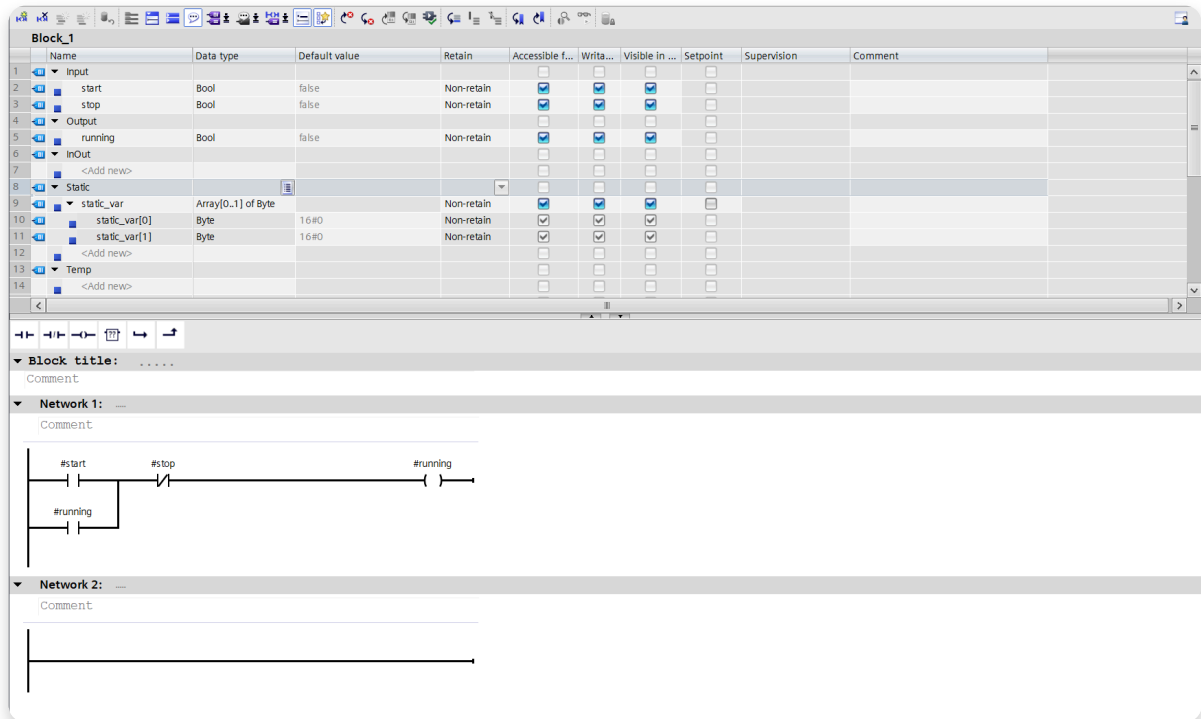


# Background

Here is a screenshot of one of the most important product in Siemens digital industry: TIA Portal.



Try to implement a simplified variable table editor that supports adding, deleting, modifying, and viewing variables.

# Core Functions

1. Table Management: Display a list of variables, including **name, data type, default value, and comment**.
2. Data Operations: Support adding rows, deleting rows, and editing cell content.
3. Data Validation: Validate uniqueness of names, data type, and default value format.
4. Support two data types: BOOL and INT (32-bit signed integer)

# Technical Requirements

- Use **React + TypeScript**
- You may use any table component library (e.g., ag-Grid, react-table, antd Table, etc.)
- You may use any CSS framework and component library
- Use any state management solution (Mobx, Redux, Zustand, etc., your choice)
- Clear code structure, clear component responsibilities
- You may use any AI assistant tools
- You may use any test framework (e.g., jest, vitest, playwright )

## Simulation Story Card

| Index | Name  | Datatype | Default Value | Comment      |
|-------|-------|----------|---------------|--------------|
| 1     | Start | BOOL     | true          | system start |
| 2     | Stop  | BOOL     | true          |              |
| 3     | Count | INT      | 123           | count number |

Please follow these ACs:

### AC 1: Table Display

- Given: the variable table editor page is opened
- When: the page is loaded
- Then:
  - Display an empty table with the following columns: Index, Name, Data Type, Default Value, Comment
  - Index column is read-only and auto-generated
  - Show "Add Row" and "Delete Row" buttons

## AC 2: Add Variable Row

- Given: the table is empty or has data
- When: the user clicks the "Add Row" button
- Then:
  - Add a new row at the end of the table
  - All fields (name, data type, default value, comment) in the new row are empty by default
  - Index is auto-generated as current max index + 1

## AC 3: Delete Variable Row

- Given: the table has at least one row
- When: the user selects a row and clicks the "Delete Row" button
- Then:
  - The row is deleted from the table
  - Indexes of subsequent rows are recalculated automatically

## AC 4: Edit Variable Name

- Given: the table has data rows
- When: the user clicks the name cell and enters a new name
- Then:
  - If the input is empty, revert to the original value and show an error prompt
  - If the input name already exists in the table (case-insensitive), show an error "Name already exists" and do not allow saving
  - If the input name is unique and not empty, save successfully

Example:

1. Table already has counter
2. User enters Counter → show error prompt
3. User enters newCounter → save successfully

## AC 5: Select Data Type

- When: the user double click the data type cell
- Then:
  - Show a dropdown with two options: BOOL and INT
  - After selection, the cell displays the selected type
- And When: the user switches the data type
- Then:
  - The default value cell is automatically updated to the default value of the new type (BOOL -> TRUE, INT -> 0)

Example:

1. Current type is BOOL, default value is TRUE
2. User changes type to INT -> default value automatically changes to 0

## AC 6: Edit BOOL Default Value

- Given: a variable row with data type BOOL
- When: the user clicks the default value cell and enters a value
- Then:
  - Accept input: true, false, TRUE, FALSE (case-insensitive)
  - Display format is unified as uppercase: TRUE or FALSE
  - If other values are entered, show an error prompt

Example:

1. User enters TRUE -> display TRUE
2. User enters false -> display FALSE
3. User enters yes -> show an error prompt

## AC 7: Edit INT Default Value

- Given: a variable row with data type INT
- When: the user clicks the default value cell and enters a value

– Then:

Accept input: integer, range -2147483648 to 2147483647

If input is not an integer or out of range, show an error prompt

Example:

1. User enters 42 -> save successfully
2. User enters 3.14 -> show an error prompt
3. User enters 9999999999 -> show an error prompt (out of range)

## AC 8: Edit Comment

– Given: the table has data rows

– When: the user clicks the comment cell and enters text

– Then:

Accept any text input

The value can be empty

## Deliverables

Source Code

1. Complete runnable project source code (remove node\_modules)
2. README.md: Include how to run the project and project structure explanation

Please upload the project to github or other code manage platform.